

MATHEMATICS
Introduction to Modern Algebra I
Quiz 8.

1) If G is a group with subgroups A and B of order m and n respectively, where m and n are relatively prime, prove that the subset of G given by $AB = \{ab|a \in A, b \in B\}$, has mn distinct elements.

Proof. Since $A \cap B$ is a subgroup of A (and of B), its order divides $m = |A|$ (and divides $n = |B|$). Since $(m, n) = 1$, we must have $|A \cap B| = 1$ i.e. $A \cap B = \{e\}$.

If $ab = a'b'$ for some $a, a' \in A$ and $b, b' \in B$, then $(a')^{-1}a = b'b^{-1} \in A \cap B = \{e\}$ and so $(a')^{-1}a = e = b'b^{-1}$. But then $a = a'e = a'(a')^{-1}a = ea = a$ and $b = eb = b'b^{-1}b = b'e = b'$.

Therefore AB has mn distinct elements.

□